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SUPERSEDING
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FEDERAL SPECIFICATION

PACKAGING AND PACKING OF TYPEWRITERS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers requirements for packaging and packing of typewriters.

1.2 Classification.

1.2.1 Options.

Option I - Detailed requirements.
Option IA - Detailed requirements (general).
Option II - Performance requirements.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

UU-T-106 - Tape, Pressure-Sensitive Adhesive, Masking, Paper.
VV-L-800 - Lubricating Oil, General Purpose, Preservative, (Water-Displacing, Low Temperature).
PPP-B-576 - Boxes, Wood, Cleated, Veneer, Paper Overlaid.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-B-640 - Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-C-843 - Cushioning Material, Cellulosic.
PPP-P-600 - Porcelain Enamel Products and Household Appliances, Electrical and Mechanical - Requirements for Packaging and Packing.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City., Mo., Ft. Worth, Denver, San Francisco, Los Angeles, and Auburn, Wash.

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Military Specifications:

MIL-P-116 - Preservation, Methods of.
MIL-L-10547 - Liners, Case, and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. Materials used for cleaning, preserving, packaging, and packing the typewriters shall conform to the applicable specifications as specified herein.

3.2 Workmanship. All operations and processes involved in accomplishing the cleaning, preservation, and unit packaging methods specified herein shall be in accordance with the highest grade practices associated with the type of work. All equipment used to perform the various procedures in accordance with this specification shall be such as to produce the specified results to the satisfaction of the procuring agency.

3.3 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

3.3.1 Level A. Unless specified otherwise, preservation and packaging shall be either option I, IA, or II at the option of the supplier (see 6.2).

3.3.1.1 Option I - detailed requirements (see 3.3.1 and 6.1).

3.3.1.1.1 Desk typewriters. All unpainted or otherwise unfinished metal surfaces of the typewriter shall be thoroughly cleaned in accordance with MIL-P-116, process C-1. Cleaned surfaces shall be thoroughly dried. These surfaces shall be coated by means of an atomized spray with preservative oil conforming to VV-L-800. Preservation shall be in accordance with MIL-P-116, method IIa. Each desk typewriter shall be processed as hereinafter specified.

3.3.1.1.1.1 Feet. The feet shall be removed from the base of the typewriter, placed in a cotton bag and securely tied to the body of the typewriter.

3.3.1.1.1.2 Ribbon spools or coverplate. The ribbon spools shall be tied to the ribbon spool cup or the coverplate secured with tape conforming to UU-T-106.

3.3.1.1.1.3 Carriage rails. Carriage rails shall be blocked by means of carriage packing clamps, carriage dowel pins, carriage brace studs or fixed wedges.

3.3.1.1.1.4 Rack release lever. Rack release lever shall be tied down to disengage the rack from the pinion. The platen latches shall be secured and the carriage centered against the left margin stop, bringing up the right margin stop and locking it in place.

3.3.1.1.1.5 Carriage. If available, manufacturer's carriage retention clamps shall be used to secure carriage in a fixed position; otherwise, carriage shall be securely tied to main frame to prevent movement.

3.3.1.1.1.6 Feed roll. Feed roll release lever shall be tied down and feed rolls released from contact with platen.

3.3.1.1.1.7 Type bars. Type bars shall be held in place with a wad of cellulosic cushioning material conforming to PPP-C-843, type II, class A or B, placed in the type bar nest.

3.3.1.1.1.8 Protrusions. All sharp protruding points on the machine which may puncture the water-vaporproof barrier material shall be protected by padding with the same type cushioning material specified in 3.3.1.1.1.7, held securely in place with tape specified in 3.3.1.1.1.2.

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3.3.1.1.1.9 Cover. The standard typewriter cover supplied with the machine shall be placed over the machine and tied securely in position or, at the option of the supplier, folded, and placed in a plastic bag which shall be sealed.

3.3.1.1.1.10 Method II procedure. To accomplish the method IIa pack (see MIL-P-116), the machine shall be securely bolted through the barrier material to the base of the box, through 3/4 inch by 3 inch interior battens which have been previously attached by clinched nails to the base of the box (see 3.4.1.1.1). The machine may either be held in an inverted position in a special packing fixture, and the water-vaporproof barrier bag placed in position over the base in order to simplify the alignment of the holes in the gaskets with the mounting points in the base of the machine, or the machine may be lowered into the barrier which has been previously laid on the cross battened wood base. The machine shall be secured by either of the methods described in 3.3.1.1.1.10.1 or 3.3.1.1.1.10.2.

3.3.1.1.1.10.1 Bolt securing. The bolts securing the machine to wooden base shall be screwed into the mounting points on the machine base through metal washers, the holes in the wooden base and battens and also through shock absorbers consisting of either metal cup and spring devices, or rubber washers of 40 Durometer hardness, minimum 1/2-inch thick by 1-1/2 inches in diameter. The bolts shall be screwed up tight. The metal washers and heads of bolts shall be countersunk into the base of the box.

3.3.1.1.1.10.2 Shock mount. Rubber shock absorbers may be used which instead of having a bolt pass through the rubber body, have threaded sections laminated to the rubber body of the shock absorber. The rubber body of the shock absorber shall be of the same material as the rubber washers. One threaded section shall be securely screwed into the typewriter frame and the opposite end securely fastened to the base of container with a flat washer, lockwasher and a hex head nut. The base of the box will be countersunk for washers and nut.

3.3.1.1.2 Portable typewriters. All unpainted or otherwise unfinshed metal surfaces of the typewriter shall be thoroughly cleaned in accordance with MIL-P-116, process C-1. Cleaned surfaces shall be coated by means of an atomized spray with preservative oil conforming to VV-L-800. Portable typewriters shall be preserved in accordance with MIL-P-116, method IIb, and in accordance with the following: Portable typewriters shall be locked in their cases with the standard locking mechanism provided. Carriages shall be blocked with dowel pins or by other suitable means. Each portable typewriter (in carrying case) shall be packaged in a corrugated fiberboard box having glued or taped manufacturer's joint (no staples shall be used). Before sealing the flaps of the box with tape or adhesive (no staples shall be used) pads of sufficient thickness, formed of folded scored fiberboard sheets, shall be placed on each side of the handle to protect the handle of the carrying case. If carrying case is not rectangular, fiberboard buffer pads shall be scored, folded, and so placed to fill void space. Application of the water-vaporproof barrier and outer box shall be in accordance with MIL-P-116, method IIb. The outer box shall conform to PPP-B-636 domestic class and shall be closed in accordance with the appendix thereto.

3.3.1.2 Option IA. Each typewriter shall be prepared in accordance with 3.3.1.1.1.2, 3.3.1.1.1.3, 3.3.1.1.1.4, 3.3.1.1.1.5, 3.3.1.1.1.6, 3.3.1.1.1.7, and 3.3.1.1.1.9. The amount of desiccant determined by the formula in MIL-P-116 for a method II pack shall be secured to the typewriter prior to placing the typewriter in a polyethylene bag, minimum 0.002 inches thick. The bag shall be sealed. A polystyrene end cap contoured on the inside for maximum support against the typewriter and the outside ribbed for maximum absorption of shock shall be placed between the polyethylene bag and the liner specified in 3.4.1.1.2.2.

3.3.1.3 Option II.

3.3.1.3.1 Preservation and packaging. All unpainted or otherwise unfinished metal surfaces of the typewriter shall be thoroughly cleaned in accordance with MIL-P-116, process C-1. Cleaned surfaces shall be thoroughly dried. These surfaces shall be coated by means of an atomized spray with preservative oil conforming to VV-L-800. All typewriters shall be further prepared in accordance with MIL-P-116, method II, submethod optional.

3.3.1.3.2 Disassembly. The extent of packaging and disassembly shall be determined by the manufacturer. This determination shall be based on the need for protection to fragile or projecting parts and for the reduction of cubage. Parts removed shall be adequately identified or tagged. Where disassembly affects the operation of a typewriter, warning tags or labels indicating the disassembly shall be attached. The instructions for assembling shall be included either as a part of the warning tag or label or as a separate sheet or folder attached to the typewriter. The disassembly shall not alter calibrations or settings which affect desired overall operation of the typewriter and shall not be such that special skills or special tools or both are required to place the typewriter in operation.

3.3.2 Level C. Typewriters shall be preserved and packaged to afford adequate protection against corrosion, deterioration, and damage during shipment from the supply source to the first receiving activity for immediate use or controlled humidity storage.

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3.4 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

3.4.1 Level A. Unless specified otherwise, packing shall be either option I or II at the discretion of the supplier.

3.4.1.1 Option I - detailed requirements (see 3.4.1 and 6.1).

3.4.1.1.1 Desk typewriters. Desk typewriters, packaged as specified in 3.3.1 or 3.3.2, shall be individually packed in a shipping container conforming to PPP-B-621, style 2, 2-1/2, or 3, or PPP-B-601, style A, except that the following special provisions shall apply.

3.4.1.1.1.1 Box components. Sides, ends, top, and bottom shall be not less than 3/4-inch thick for PPP-B-621 and 1/4-inch thick for PPP-B-601.

3.4.1.1.1.2 Cleats. Cleats shall be not less than 3/4-inch thick and 2-1/4 inches wide for PPP-B-621 and 1-3/16 inches thick and 2-3/8 inches wide for PPP-B-601. Interior battens shall be not less than 3/4 inch thick and 3 inches wide.

3.4.1.1.1.3 Bottom. The bottom of each box to which the unit is bolted shall be fastened to the sides and ends of the box with not less than 1-3/4 inch No. 8 flat head wood screws, spaced not more than 3 inches apart on the ends and 5 inches apart on the sides.

3.4.1.1.1.4 Strapping. Shipping containers shall be strapped in accordance with the appendix of PPP-B-621 or PPP-B-601.

3.4.1.1.1.5 Size. Standard size shipping containers conforming to the dimensions shown in table I shall be used for the packing of desk typewriters.

TABLE I. Shipping container sizes for desk typewriters

Description of typewriters	Inside dimensions of shipping container		
	Length	Width	Depth
	Inches	Inches	Inches
10, 11, and 12 inch	21-1/2	18-1/2	12
14 inch	23	18-1/2	12
18 inch	28	18-1/2	12
20, 22, and 26 inch (except 26 inch Underwood)	37	18-1/2	12
32 inch	1/	18-1/2	12

1/ The length of the shipping container shall be determined by adding two inches to the actual carriage or carriage rail length, whichever is greater.

3.4.1.1.2 Portable typewriters. Two portable typewriters, packages as specified in 3.3.1 or 3.3.2, shall be packed flat in a shipping container conforming to PPP-B-621, style 2, 2-1/2, or 3, or PPP-B-601, overseas type, style A.

3.4.1.1.2.1 Strapping. Shipping containers shall be strapped in accordance with the appendix of PPP-B-621 or PPP-B-601, as applicable.

3.4.1.1.2.2 Case liner. Each shipping container shall be provided with a type I or II, grade C case liner conforming to MIL-L-10547.

3.4.1.2 Option II. One desk or two portable typewriters, packaged as specified in 3.3.1 or 3.3.2, shall be packed in a manner that shall meet the test requirements of 4.2.2. The shipping container shall be considered to have satisfactorily passed the tests if, upon examination, the typewriter is free from damage and the container and inner packing have only superficial scuffs, scratches, or dents. The shipping container shall be constructed, closed, waterproofed and strapped in accordance with one of the following: PPP-B-585, class 3; PPP-B-591, class II, style A or B; PPP-B-601, overseas type, style A or B; PPP-B-621, class 2; PPP-B-636, V3c or V3s, style optional; PPP-B-640, class 2; or PPP-B-576, class 2. All packaging and packing materials subject to water damage shall be of water-resistant types. Containers not already waterproofed shall be provided with a case liner as specified in 3.4.1.1.2.2.

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3.4.2 Level B. Packing shall be option II only. The typewriter, packaged as specified in 3.3.1 or 3.3.2, shall be packed in a manner that shall meet the test requirements of 4.2.3. There shall be one desk or two portable typewriters in each shipping container. The unit pack shall be considered to have satsifactorily passed the tests if, upon examination, the typewriter is free from damage and the container and inner packing have only superficial scuffs, scratches, or dents. Shipping containers shall conform to the following:

a. PPP-B-636, class domestic, variety SW or DW, type CF or SF.
b. PPP-B-591, class I.
c. PPP-B-640, class 2.
d. PPP-B-601, domestic type.
e. PPP-B-576, class 1, style A or B, type 3 load.
f. A molded plastic container of expanded polystyrene or equivalent material.

3.4.3 Level C. Packing shall be option II only. The typewriters, packaged as specified in 3.3.1 or 3.3.2, shall be packed in a manner that shall meet the test requirements of 4.2.4. There shall be one desk or two portable typewriters in each shipping container. The unit pack shall be considered to have satisfactorily passed the tests, if, upon examination, the typewriter is free from damage and the container still affords reasonable and proper protection to the contents. The packing materials shall be those materials which are normally used by the supplier for shipment of the product.

3.5 Marking.

3.5.1 Civil agencies. In addition to any special markings specified in the contract or order, marking shall be in accordance with Fed. Std. No. 123.

3.5.2 Military agencies. In addition to any special markings required by the contract or order, marking shall be in accordance with MIL-STD-129.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated herein.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.2.1.1 Examination of preservation. Examination of preservation shall be in accordance with the sampling procedures, inspection level and acceptable quality levels (AQL) as specified in MIL-P-116.

4.2.1.2 Examination of the end item for packaging, packing, and marking. The end items shall be examined in accordance with the classification of defects in table II. The sample unit shall be one desk or two portable typewriters as applicable, fully prepared for shipment. The lot shall be expressed in terms of typewriters fully prepared for shipment. The inspection level shall be S-2 and the acceptable quality levels (AQLs) shall be 2.5 for major and 10.0 for total defects expressed as defects per hundred units.

TABLE II. Examination for packaging, packing, and marking

Examine	Defect	Category	
		Major	Minor
Packaging			
Level A, options I and II.			
Desk typewriters			
Feet	Feet not removed when required.	101	
	Feet not placed in cloth bag and securely tied to body of typewriter when required.		201

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TABLE II. Examination for packaging, packing, and marking (cont'd)

Examine	Defect	Category	
		Major	Minor
Packaging (cont'd)			
Level A, options I and IA. (cont'd)			
Desk typewriters (cont'd)			
Ribbon spools	Not tied to spool cup.		202
Coverplate	Not taped securely.		203
Carriage rails	Not blocked by one of specified methods.	102	
Rack release lever	Not tied down.	103	
	Platen latches not secured.		204
Carriage	Carriage not centered on machine with stops securing it in place.	104	
Feed roll	Lever not tied down.		205
	Rolls contacting platen.		206
Type bars	Cellulosic cushioning missing or not properly placed.	105	
Protrusions	Cushioning when required missing or inadequate to prevent puncturing. Not taped in place.	106	
Cover	Missing, not securely tied in position, nor in sealed plastic bag.		207
Bolting method	Bolts not tightened securely.	107	
	Shock absorbers, washers missing.	108	
	Bolt heads not countersunk into base of box.	109	
Portable typewriters	Not locked in case with locking mechanism provided.	110	
	Carriage not adequately blocked to prevent damage to rails.	111	
	Carrying case not closed by method provided.		208
	Fiberboard carton missing.	112	
	Scored and folded pads missing when required to fill voids.	113	
Level A, option II.			
Disassembly	Identification tags missing from disassembled parts.		209
	Instruction tags missing when required for reassembly.	114	
	Calibrations affected by disassembly.	115	
Packing			
Level A, option I			
Shipping container			
Box components (cleats, ends, battens)	Width and thickness less than specified.	116	

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TABLE II. Examination for packaging, packing, and marking (cont'd)

Examine	Defect	Category	
		Major	Minor
Packing (cont'd)			
Level A, option I (cont'd)			
Shipping container (cont'd)			
Bottom (desk typewriter only)	Not fastened to sides and ends with screws.	117	
	Screws spaced more than 3 inches apart on ends or 5 inches apart on sides.		210
	Not No. 8 x 1-3/4 screw.	118	
	Container sizes not in accordance with table I.	119	
	Not individually packed in box (desk typewriter only).	120	
	Not packed two each flat on bottom in box (portable typewriter only).	121	
	Case liner missing when specified.	122	
Marking	Omitted, incomplete, incorrect, illegible, of improper size, location, sequence or method of application	123	

4.2.1.3 Examination of shipping containers. When shipping containers are required to be in accordance with PPP-B-576, PPP-B-585, PPP-B-591, PPP-B-601, PPP-B-621, PPP-B-636, or PPP-B-640, examination for closure and strapping shall be in accordance with the applicable appendix of those specifications.

4.2.2 Level A, option II. Examination and testing shall be in accordance with level A requirements of PPP-P-600 for vibration, impact, compression, and drop.

4.2.3 Level B, option II. Examination and testing shall be in accordance with level B requirements of PPP-P-600 for vibration, impact, compression, and drop.

4.2.4 Level C, option II. Examination and testing shall be in accordance with level C requirements of PPP-P-600 for vibration, impact, compression, and drop.

5. PREPARATION FOR DELIVERY

(Not applicable)

6. NOTES

6.1 Intended use. The preservation, packaging, and packing requirements of this specification are for the preparation of all types of typewriters for delivery in accordance with any of the following options (see 3.3.1, 3.4.1, 3.4.2, and 3.4.3): Options I and IA - detailed requirements for preservation and packaging, and option I for packing of typewriters. Option II - performance requirements for preservation, packaging, and packing of typewriters.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Whether option I, IA, or II is desired (see 3.3.1 and 3.4.1).
(c) Level of preservation and packaging (see 3.3).
(d) Level of packing (see 3.4).
(e) Special markings required (see 3.5).

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